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NLP 201: Natural Language Processing I

Problem 1:

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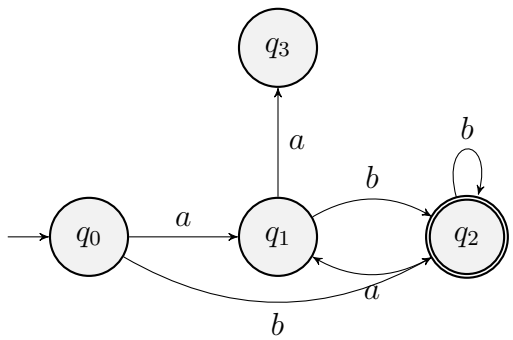

Figure 1: Problem 1 DFA

Problem 2:
2.1

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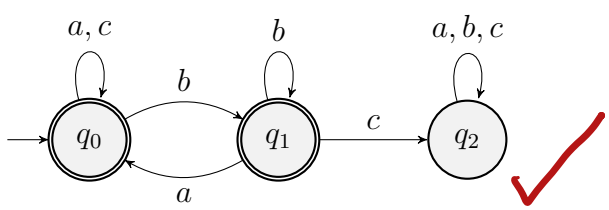

Figure 2: Problem 2.1 DFA

2.2

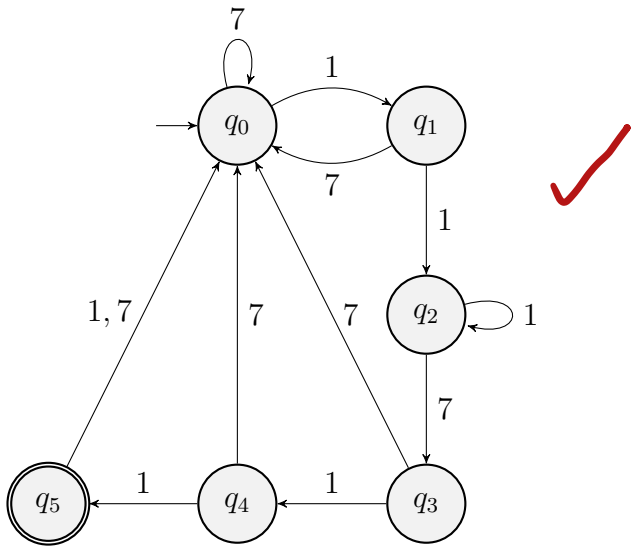

Figure 3: Problem 2.2 DFA

Problem 3:

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If q0 get 1 and go to q1 if the input is 7 how to deal with the problem, using the diagram.

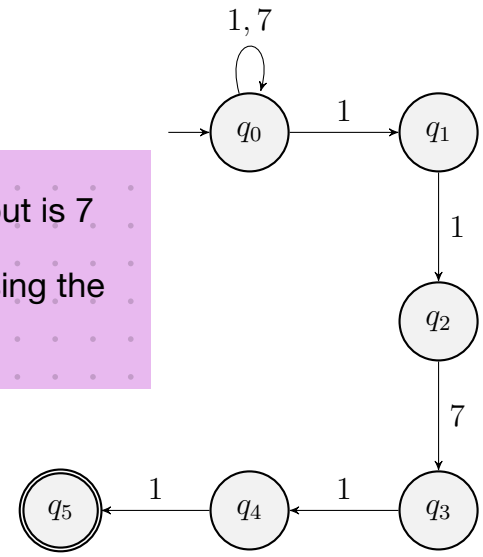

Figure 4: Problem 3 NFA

There is no transition $\delta(q_i, 7)$ for state q_1, q_3 and no self loop on state q_2 since those can be contained in the self loop at the start state for NFA. There's no need to have trash state after the accepting state since NFA rejects alphabet if there aren't ant valid transition.

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Problem 4:

4.1 $(1|7)^*711$ ✓

4.2 $(a|b|c)^*(ab|bc)(a|b|c)^*$ ✓

4.3 $(a(a|b|c)^*ab(a|b|c)^*)|(b(a|b|c)^*bc(a|b|c)^*)$ Do we need add (a|b|c)

Problem 5:

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Proof. We've proved in class that $\emptyset$, which is empty language, is regular in class. We've also proved that ϵ, which is the language $\{\epsilon\}$, is regular. We've also proved that σ, which is the language $\{\sigma\}$, is regular. $(\alpha|\beta)$ is regular since it is a GRE denoting $A \cup B$ and regular languages are closed under union. $(\alpha\beta)$ is regular since it is a GRE denoting $A.B$. And regular languages are closed under concatenation which means that $A.B$ is regular.

Now we need to prove that regular languages are closed under union.

Proof. Let L_1 and L_2 be two regular languages. Then there must exists two DFAs M_1, M_2 such that $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1), M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ and M_1 recognizes L_1, M_2 recognizes L_2.
We can construct a new DFA $M_3 = (Q, \Sigma, \delta, q_0, F_0)$ which recognizes L_3, and $L = L_1 \cup L_2$.
M_3 will be pairs of states from both M_1 and M_2 which means that each state in Q_3 is (q_1, q_2) pair and $q_1 \in Q_1, q_2 \in Q_2$.
Alphabets are the same for both machines therefore it's the same for L_3.
The start state q_0 will be the paired state of start states from both machines which would be (q_1, q_2).
The transition function instead of $\delta(q_i, \sigma)$ will be $\delta((q_1, q_2), \sigma) = (\delta_1(q_1, \sigma), \delta_2(q_2, \sigma))$, which indicates how both machines will transit on the same letter.
And the accept states will be a paired state of accept states of both machines which will be (q_i, q_j) and $q_i \in F_1, q_j \in F_2$.
In this way the machine M_3 will be able to accept any strings that can be accepted by either M_1 or M_2, which means that the machine M_3 recognizes $L = L_1 \cup L_2$. And thus we can conclude that $L1 \cup L_2$ is a regular languages.
Therefore regular languages are closed under union. □

We also need to prove that regular languages are closed under complement.

Proof. Let L be a regular language. Then there must exists a DFA M such that $M = (Q, \Sigma, \delta, q_0, F)$ and M recognizes L.
We can construct a new DFA $M_1 = (Q, \Sigma.\delta, q_0, Q \backslash F)$ for $\overline{L}$.
This machine is almost the same as M but all the accpet states in M will be non-accepting states in M_1. And all non-accepting states in M will be accepting states in M_1. In this way, M_1 will except any languages but L, which is $\overline{L}$. Since machine M_1 is a DFA and it recognizes $\overline{L}$, it means that $\overline{L}$ is a regular language. Therefore, regular languages are closed under complement. □

Since we've proved that all GREs operations preserve regularity in this problem. We can conlude that any languages denoted by GREs are regular. □

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Problem 6:

6.1 False. A counter example would be $L_2 = \Sigma^*$ where $\Sigma = \{a, b\}$ and $L_1 = \{a^n b^n | n \geq 1\}$. ✓

6.2 False. A counter example would be $L_1 = \phi$ which is regular, thus $L = L_1 = \phi$, which does not imply that L_2 is regular.

6.3 False. If $L_1 = L = \Sigma^*$, then it doesn't matter whether L_2 is regular or not.

6.4 False. If $L_1 = \{\Sigma^* \backslash \{a^1 b^1\}\}$, $L_2 = \{\Sigma^* \backslash \{a^2 b^2\}\}$, $L_i = \{\Sigma^* \backslash \{a^i b^i\}\}$. Then the intersection $L = \{\Sigma^* \backslash \{a^n b^n\}\}$. However, the compliment of L is $\{a^n b^n | n \geq 0\}$. Since we've proved that regular languages are closed under compliment, therefore L is not regular.

6.5 True. Since β and $\alpha\gamma$ are contained within $L(\gamma)$, and $\alpha^*\beta$ means that any number of α followed by β, which are also included in $L(\gamma)$.

6.2 We can not just prove $L_1 = \emptyset$

$L_1 = \{a, b\}$ $L = \{ab\}$ $L_2 = \{a^n b^n$

6.3 $L = L_1 \cup L_2 = \Sigma^*$.